PIC – Inventory Module Analysis and Completion

Date: 2 November 2026. Written for: the product owner, as the record of what the specification requires of inventory, what was found, what was built and fixed, how it was tested, and what is still open.

1. Finding in one paragraph

The Technical Document (TD) and Blueprint (BP) do not define a stand-alone inventory or warehouse module. Stock is a receipt-driven ledger that sits inside the supplier-fulfilment module (BP §15, M09), the special-stock rules (BP §22–23), the MRP supply formula (TD §9.1) and a handful of KPIs. Before this work the ledger was written correctly by goods receipt, quality decisions, consignment and subcontracting, but **nothing could read it and nothing governed changes to it**: there were no stock, movement, summary or alert endpoints, no screens, and no controlled way to post a transfer, block, scrap, count or reservation. The reservation demand type existed in the planning model but nothing created it. The old `/inventory` screens in the web app are a separate prototype with their own mock tables; they were left alone.

2. What the specification requires, and the state found

#	Requirement (source)	State before	State now
1	Goods receipt updates stock, GR/IR and scorecard (BP F6)	Built	Unchanged, defects fixed (section 3)
2	GR for quality-managed materials posts to quality stock until usage decision (BP §15.4)	Built	Built; cut-off check added
3	Batch, serial, expiry capture where required (BP §15.1)	Built, with a serial-reuse defect and a 500 on bad dates	Fixed
4	Over-delivery above tolerance blocked at receipt (BP §15.4, TD §5.2)	Built	Unchanged
5	Consignment and subcontracting stock (BP §15.1, §22)	Built, no plant check	Plant and period checks added
6	MRP supply uses unrestricted on-hand (TD §9.1)	Built	Unchanged (consignment decision open, section 6)
7	Reservations are gross demand in MRP (TD §9.1)	Demand type only, nothing created it	Reservations create and release the demand
8	Stock/requirements list per material (BP MRP-02)	Missing	Material stock card with per-bucket projection
9	KPIs: stock-outs, days of cover, dock-to-stock, quality rejection (BP)	Built	Stock-out definition corrected; inventory turns, count accuracy, adjustment value added
10	Expiry write-offs, par levels, cold chain (healthcare template)	Built, expiry sweep wrongly used reversed receipts	Fixed
11	Org hierarchy plant → storage location, deactivated sites refused (BP §4.1)	Built for receipts	Applied to all stock postings
12	Segregation of duties, audit, ERP posting, idempotency (TD §3, §10)	Built for receipts	Applied to all new inventory commands
13	Read access to stock, movements, alerts (implied by every screen that shows stock)	Missing	Built
14	Governed stock adjustments (transfer, block, scrap, count)	Missing	Built (not named in the specification; added because the old prototype offered adjustments and an enterprise tenant needs them controlled)

3. Defects found and fixed

Each has a regression test in `backend/tests/pic/test_inventory_ledger.py`.

1. **Serial numbers of a reversed receipt could never be received again.** The duplicate-serial check included reversed receipts, so the corrected receipt was refused. It is now one indexed query that ignores reversed receipts.
2. **A malformed expiry date returned HTTP 500.** Expiry and ASN batch dates are now validated and return 422.
3. **Reversing a receipt and taking a usage decision skipped the closed posting period.** Both now apply the cut-off.
4. **Missing indexes and constraints.** Ledger lookups had no supporting index. Added five indexes, a stock-type domain check and a non-negative quantity check (after confirming no seed or ERP-sync path writes a negative balance).
5. **Expiry sweep and value-analysis report used reversed receipts**, which could condemn good stock.
6. **Consignment withdrawal, recall, subcontractor components, expiry write-off, usage decision and receipt reversal had no plant (ABAC) check** and mostly no period check. Added.
7. **Quantities with more than three decimals were silently rounded** by the database; wrong-sign movements were accepted; two first postings of a balance could race. All now refused or serialised.
8. **Opening stock in two seed tenants had no ledger movement**, which would have shown as false drift. The seeds now post an opening movement.
9. **The stock-out KPI counted non-stock materials (services, expense items) and ignored the plant on the demand side.** Corrected in this pass.

4. What was built

- **Ledger service** (`stock_ledger.py`): one posting primitive used by receiving and the new code, with row lock, negative-stock guard, value at standard price, reason code and document reference on every movement.
- **Reads** (`stock_reads.py` , `/v1/stock…`): filterable stock list, summary (value, quality hold ageing, blocked, expiring, below safety, stock-outs, days of cover, open reservations and counts), typed alerts (stock-out, below safety, below reorder, expiring, expired, quality hold aged, blocked aged, drift), material stock card with the MRP-02 requirements profile, movement ledger and a capped, hashed, audited CSV export.
- **Stock postings** (`STP-`): transfer, block, unblock, scrap. Draft, submitted, approved, posted, rejected, cancelled. Value-banded approval chain; small transfers, blocks and unblocks post automatically; scrap always needs approval; the maker can never approve (SOD-10, enforced per document). Stock is re-checked under lock at posting.
- **Physical inventory** (`PHI-`): count freeze, blind counting, tolerance-driven recount by a different person, approval by someone who did not count (SOD-11), differences posted as 701/702 relative to the freeze so receipts during counting are not wiped out.
- **Reservations** (`RSV-`): availability check, demand for MRP, partial issue (first-expiry-first-out), cancel, daily expiry job.
- **Rule sets** (effective-dated, editable through the admin rules route): `INVENTORY-POSTING` , `INVENTORY-COUNT` , `INVENTORY-RESERVATION` , `INVENTORY-ALERTS` .
- **Access**: scopes `inventory:post` , `inventory:count` , `inventory:approve` ; new built-in role `INVENTORY_CONTROLLER` ; Receiver may post and count; Finance Controller may approve.
- **Integration**: ERP message type `STOCK_POSTING` , idempotent per document; 8 event schemas; KPIs `INVENTORY_TURNS` , `STOCK_COUNT_ACCURACY` , `STOCK_ADJUSTMENT_VALUE` .
- **Data**: migration 0018; demo pack `inventory_ops` for the AXN and KAC tenants (KAC also receives an opening position of 20 materials).
- **Screens**: Inventory (stock, alerts, movements, postings, counts, reservations), material stock card, and detail pages for postings, counts and reservations with the usual approvals, documents, lineage and audit tabs; navigation entry; links from receipts and lot trace.
- **Runbook**: `docs/runbooks/inventory_operations.md` .

5. Test evidence

- Inventory ledger tests 14, inventory journey 21 steps through the HTTP API, inventory demo pack 5: passing.
- Dataset, industry and module-case tests (62) passed on a fresh database. One apparent failure in the KAC test came from running the inventory pack first in the same database; it passes in the normal order.
- About 765 of 849 backend tests ran in the engineer's full run with no pre-existing test failing; the full suite has **not** been run to completion on the final code.
- Migration 0018: upgrade, downgrade to 0017, upgrade again and model/migration comparison all passed on a scratch database.
- Browser (real flows against a scratch database): every tab and the stock card as controller, receiver and plain reader; a transfer that auto-posts; a scrap that waits for approval, refused for its maker and approved by a second controller; a blind count entered by a receiver, approved and posted by a controller with a loss; a reservation issued in two parts and one cancelled; CSV export; 390 px mobile width; keyboard tab navigation. No console errors or failed calls.
- Frontend `tsc` exit 0; ESLint on the CI scope 0 errors.

6. Open items and decisions

1. **Not run in the browser:** block and unblock postings, rejecting a posting, a Finance Controller approving, a controller requesting a recount, and the batch, material-group, expiring and below-safety filters. The shortage highlighting in the material card was not seen because the demo data has no demand inside the 90-day horizon.
2. **Full backend suite** to be run once on the final code before the push.
3. **Decision: consignment stock in MRP supply.** The TD says unrestricted only, so consignment goods are neither open order nor on-hand supply after receipt. Left as specified.
4. **Decision: cancelled or issued reservations** set the demand to zero and keep the row so net-change MRP replans the material; the daily job removes zero rows after seven days.
5. **SOD-10** is enforced per document, not as a role-pair rule, because the new controller role legitimately holds both posting and approval. Consequence: a Receiver cannot also be an Inventory Controller or Finance Controller.
6. **Known, not changed:** the ERP stock sync writes an empty-batch balance that can double-count batch-managed materials; fractional quantities are accepted for each-unit materials (no unit-of-measure rule); demo documents are dated 2027 by the pinned business clock, so new documents sort below them in lists.
7. The old `/inventory` prototype screens still use their own mock tables and are not part of the PIC lint scope.

7. What the product owner must supply

- ERP mapping of the `STOCK_POSTING` message (goods-movement BAPI or material-document API) for any live ERP; the SAP simulator works today.
- Standard prices, safety stock, reorder points and batch/expiry flags on materials; opening stock for real tenants (ledger opening movement or ERP stock sync).
- At least two Inventory Controller users per plant, with plant and value-limit attributes.
- Decisions on the rule values: auto-post limit (default ₹25,000), approval bands, count tolerance (2% and ₹5,000), scrap reason codes, reservation grace days (7), alert ageing limits.
- A decision window to apply migration 0018 to production, then the normal Neon setup run (it seeds number ranges, rule sets, SoD rules and the demo pack).